

Auteur: Abdul Ahmed
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$$\begin{aligned} & \frac{d}{dx} ((\sin x)^{\cos x}) \\ &= \frac{d}{dx} (e^{\cos x \cdot \ln(\sin x)}) \\ &= e^{\cos x \cdot \ln(\sin x)} \left(-\sin x \cdot \ln(\sin x) + \frac{\cos x}{\sin x} \cdot \cos x \right) \\ &= (\sin x)^{\cos x} (\cot x \cdot \cos x - \sin x \cdot \ln(\sin x)) \end{aligned}$$

References